DiffBench: Toward Robust and Reproducible Prediction in Diffusion MRI

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Abstract. Diffusion MRI is widely used to explore brain microstructure, and numerous machine learning methods have been proposed to extract predictive information. Yet progress remains difficult to assess: studies rely on single datasets, heterogeneous pipelines, limited evaluation, and often lack accessible code, preventing meaningful comparison and reproducibility. We introduce DiffBench, a modular and transparent benchmarking framework for machine and deep learning in dMRI microstructure. The benchmark unifies multiple representative datasets within a standardized end-to-end pipeline and enables systematic comparison across tissue types, feature representations and models. It reveals differences between white and gray matter analyses, quantify the impact of diffusion microstructure feature choices, and show that fast linear baselines can rival more complex deep learning methods. This shows that standardized benchmarking is essential to move the field beyond proxy tasks and toward robust methods with real clinical relevance.

Keywords: Benchmark · Brain · diffusion MRI · Machine Learning · Deep Learning · Predictive Learning · Reproducibility.

1 Introduction

Diffusion magnetic resonance imaging (dMRI) enables in vivo representation of tissue microstructure and provides biologically meaningful biomarkers sensitive to axonal organization, myelination, and tissue complexity [29]. Leveraging these microstructural descriptors for predictive modeling has become increasingly common, with applications ranging from demographic estimation [7, 22, 5, 27, 15], clinical diagnosis [16, 40], and cognitive characterization [25, 26]. The combination of diffusion imaging and machine learning is therefore often viewed as a powerful framework for data-driven neurobiological inference.

While many studies demonstrate the promising predictive power of dMRI, reported performance varies widely across studies, tasks, and datasets, making direct comparisons difficult. This variability arises from multiple sources. Preprocessing pipelines differ substantially, microstructural information is represented in diverse ways ranging from voxel-wise maps [7] and ROI summaries [26] to learned embeddings [6, 22] and evaluation protocols vary in data splits, cross-validation schemes, and hyperparameter selection. Limited code and configuration sharing further restrict reproducibility. Consequently, reported performance

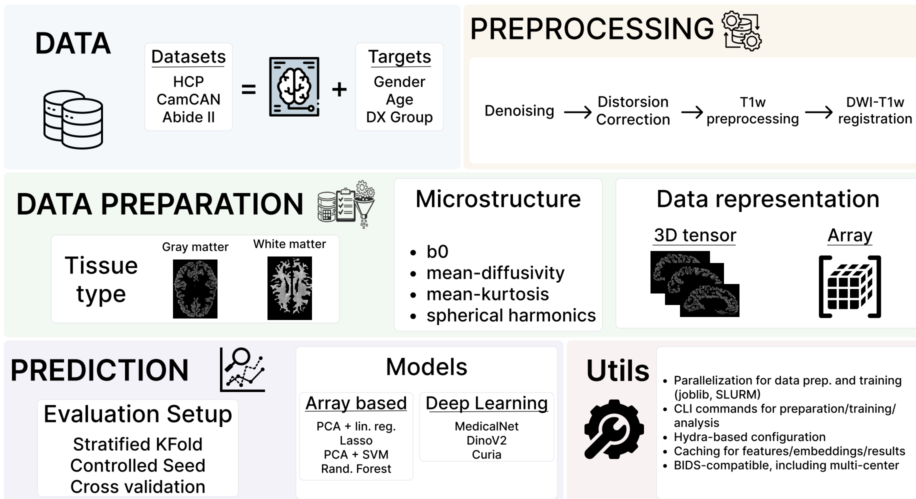

Fig. 1. Overview of the proposed diffusion MRI benchmark pipeline end-to-end.

differences may reflect inconsistencies in implementation and evaluation rather than genuine modeling improvements.

Addressing replication challenges in brain structure–behavior research [41, 19, 32] therefore requires control of these methodological sources of variability and comparable evaluation standards for dMRI prediction. Accordingly, there has been increasing interest in reproducible benchmarking frameworks within specific clinical contexts or imaging modalities [40], and large-scale analyses have assessed the replicability of dMRI-based brain–behavior associations across phenotypes [32]. However, these efforts do not jointly vary microstructural representations, tissue compartments, model families, datasets, and predictive tasks within a unified experimental design. As a result, several fundamental questions remain unsolved: how representation and model choice interact with task characteristics, whether white and gray matter contribute differently across datasets and prediction targets, and under what conditions diffusion-derived features provide consistent predictive value beyond non-diffusion baselines.

To address this, we introduce an open-source dMRI benchmark that enables controlled comparison across datasets, tissue types, microstructural representations, and model families. The benchmark explicitly quantifies performance variability across folds and configurations, isolates representation and tissue-specific effects, and evaluates the value of diffusion-derived features beyond non-diffusion baselines. We believe this provides the foundation for isolating microstructural signal from configuration-driven variability and for establishing comparable evaluation standards in diffusion MRI prediction.

2 Benchmark Pipeline

In this work, we introduce **DiffBench** a standardized benchmark that unifies preprocessing and feature extraction, evaluates both classical and deep learning models on regression and classification tasks over a rigorous evaluation protocol on multiple public datasets. The overall structure is illustrated in Figure 1.

2.1 Datasets

To ensure reproducibility, DiffBench leverages three publicly available datasets covering both healthy and clinical populations. **HCP**: The Human Connectome Project (HCP) Young Adult dataset [37] targets gender and age. It includes 1,036 healthy subjects after filtering (474 males, 562 females; ages 22-37). HCP data were acquired on a customized Siemens 3T Connectome scanner using a multi-shell Spin-Echo EPI sequence (TR/TE = 5520/89.5 ms, 1.25 mm isotropic, multiband factor 3; b-values = 1000, 2000, 3000 s/mm^2, 90 directions/shell). **Cam-CAN**: The Cambridge Centre for Ageing and Neuroscience (Cam-CAN) dataset [33] targets gender and age. It includes 629 healthy subjects (316 males, 313 females; ages 18-88), scanned on a Siemens 3T TIM Trio using a 2D twice-refocused SE EPI sequence (TR/TE = 9100/104 ms, 2.0 mm isotropic; b-values = 1000, 2000 s/mm^2, 30 directions each). **ABIDE II**: The Autism Brain Imaging Data Exchange II (ABIDE II) dataset [14] targets Autism Spectrum Disorder (ASD) diagnosis. It includes 218 subjects (186 males, 32 females; ages 5-64), including 119 with ASD and 99 controls. As a multi-site dataset, ABIDE II data were acquired on various 3T scanners with highly heterogeneous, site-specific single-shell diffusion protocols. Across the included sites, spatial resolutions range from 0.94 to 3.0 mm, TR/TE from 5200-20244/78-101 ms, b-values include 1000, 1500, and 2500 s/mm^2, and diffusion directions vary between 32 and 64. This variability provides a realistic scenario for evaluating model robustness to scanner heterogeneity.

2.2 DiffBench Preprocessing

Except for HCP, which is already preprocessed, dMRI data are preprocessed using the `anonymized` package[1] which containerises various tools such as `dipy` [18], `sMRIPrep` [17], `FSL` [35], `ANTs` [2], `FreeSurfer` [11] and `AFNI` [10]. The images are first denoised with the Marchenko-Pastur PCA (MPPCA) method [39, 9] implemented in `dipy`. This denoising procedure identifies the principal components corresponding to noise and then removes them from the dMRI data [39]. After MPPCA denoising, Gibbs ringing artifacts [38] are also removed with `dipy`. Next, we correct the distortions due to inhomogeneities of the magnetic field with `FSL`'s eddy correction [1]. Finally, this denoised and distortion-corrected dMRI data are registered to the anatomical T1w image preprocessed with `sMRIPrep`, using `FreeSurfer`'s *bbregister* [21] with 12 degrees of freedom and boundary-based registration cost function.

[1] anonymized link

2.3 Data Preparation

Microstructural representation To characterize the diffusion signal, various microstructural measures from established diffusion models are implemented in DiffBench. These measures capture complementary aspects of tissue architecture and have been widely used in previous studies [23, 22, 6, 4], ranging from diffusion magnitude to more complex descriptors:

- **Baseline ($b = 0$):** The averaged non-diffusion-weighted (T2w) signal is included as a reference to disentangle diffusion-driven predictive effects from underlying structural T2 contrast.
- **Mean Diffusivity (MD):** MD is computed via Diffusion Tensor Imaging (DTI) [3], and reflects the overall magnitude of water diffusion.
- **Mean Kurtosis (MK):** Mk is derived from Diffusion Kurtosis Imaging (DKI) [24], quantifies deviations from Gaussian diffusion and acts as a proxy for microstructural heterogeneity and tissue complexity.
- **Spherical Harmonics (SH) Power:** SH is a model-free representation obtained by computing the L2 norm of the SH coefficients [36] fitted to the diffusion signal (up to order 6). This representation especially captures signal anisotropy without assuming a specific compartmental model.

To mitigate free-water effects and scanner-dependent intensity scaling [31], we apply ventricular normalization: for each subject and each microstructural representation, voxel-wise diffusion measures are divided by the mean value extracted from cerebrospinal fluid within the ventricles. Finally, extreme values are clipped at the 99th percentile to ensure numerical stability during model training.

Spatial normalization We apply distinct spatial normalization strategies for gray and white matter to enable anatomically consistent comparisons across subjects and datasets while respecting tissue-specific geometry.

Gray Matter. Gray matter microstructure is represented on the cortical surface reconstructed from T1w images with FreeSurfer. Cortical meshes (white and pial surfaces) are parcellated with the Desikan–Killiany atlas [13]. Diffusion scalar maps are computed in the native diffusion space and rigidly aligned to the anatomical image. Voxelwise diffusion measures are then projected onto the individual cortical surface with ribbon-constrained sampling, preserving vertex-wise resolution ($\sim$ 64k vertices per hemisphere) [21]. This surface-based strategy respects cortical geometry, reduces partial-volume effect, maintains high spatial specificity and enables consistent spatial normalization across subjects.

White Matter. White matter microstructure is represented with Tract-Based Spatial Statistics (TBSS) [34]. Diffusion scalar maps are nonlinearly registered to standard space and projected onto a group-wise white matter skeleton, which captures the core of major fiber bundles and minimizes residual misalignment and partial-volume effects. Then, we summarize microstructural valuse by extracting the mean for 48 major tracts from the JHU White-Matter Tractography Atlas [28]. This normalization enforces anatomical correspondence across subjects and reduces variability associated with subject-specific tractography.

While this approach sacrifices voxel-level granularity compared to the cortical surface representation, it provides a compact, reproducible, and anatomically interpretable feature set that is robust in multi-site settings.

2.4 Prediction and validation

BenchDiff includes a broad range of predictive models previously used in dMRI studies [40, 4], from classical machine learning pipelines to foundation models. Classical models (Ridge, Lasso, Random Forests, and SVM), implemented in `scikit-learn`, can be combined with PCA for dimensionality reduction [26]. Recent foundation models, DINOv2 [30] and Curia [12], implemented in `PyTorch`, are used as frozen feature extractors with task-specific prediction heads trained on top. These models operate on 2D slices extracted along the x-axis. In contrast, MedicalNet [8] is applied to full 3D diffusion volumes, initialized from pretrained weights and fine-tuned end-to-end. Deep models are trained using AdamW, with exponential learning-rate scheduling, dropout, and weight decay. Hyperparameters are selected via cross-validation. Details are fully available in the codebase.

The prediction tasks covers classification and regression and were selected based on data availability and existing literature. **Classification.** BenchDiff includes binary classification tasks, such as Autism Spectrum Disorder (ASD) diagnosis (ASD vs. Typically Developing) using ABIDE II [4], and gender prediction in the HCP and Cam-CAN cohorts [7, 15]. Performance is reported using balanced accuracy, which accounts for class imbalance. **Regression.** Brain age prediction [5, 27, 23] was used as a regression task on the healthy Cam-CAN cohort. Performance is quantified with R^2 and mean absolute error (MAE).

We adopt a stratified 5-fold cross-validation scheme for all tasks. For classification, stratification ensures class balance across folds. For regression, splits are stratified by gender. Random seeds are fixed to guarantee reproducibility.

2.5 Benchmark implementation

To ensure reproducibility and fair comparison, the benchmark is implemented as an open-source, modular pipeline. Configuration is handled with `Hydra`[2], enabling systematic control of data processing, training, and evaluation. Data are organized according to the BIDS standard [20]. The framework supports parallel execution for data preparation and training with SLURM, and caches intermediate results to ensure efficient and reproducible experimentation.

3 Results

We evaluated the predictive power of dMRI microstructural representations across datasets, tasks, tissue types, and model families to assess performance

[2] https://github.com/facebookresearch/hydra

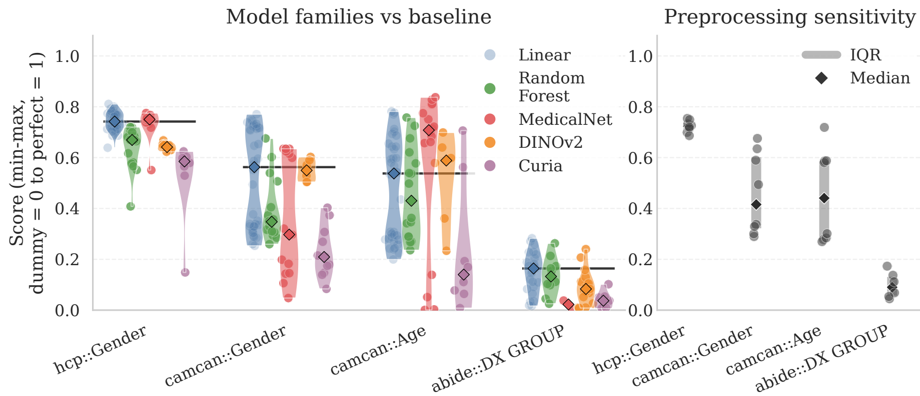

Fig. 2. *(Left)* Model-family performance normalized across metrics and folds (dummy = 0, perfect = 1) reveals no consistent advantage of any model class. *(Right)* Preprocessing sensitivity across (feature, tissue) configurations demonstrates that performance strongly depends on representation and tissue selection.

stability, tissue specific contributions, and the added value of diffusion-derived microstructural features beyond non-diffusion baselines.

Figure 2 highlights two central findings. First we evaluated the relative performance of different model families across datasets and configurations. To enable direct comparison across tasks, performance metrics were rescaled using min–max normalization. The results show that no single model consistently outperforms the others, and that regularized linear models remain competitive with deep learning architectures. This suggests that current data representations may not fully exploit the representational capacity of deep models, or that they are not sufficiently informative to justify increased model complexity. In the context of dMRI, limited sample sizes combined with high-dimensional signals further constrain the effective use of deep architectures – e.g. ABIDE II includes only 218 samples. Indeed, Curia and DINOv2 were employed as frozen feature extractors without fine-tuning, a transfer-learning regime that typically requires larger cohorts to achieve competitive performance.

Second, preprocessing and microstructural choices introduce substantial variability. In several tasks, differences across feature and tissue configurations exceed those between model families. This indicates that performance cannot be attributed to model choice alone. Preprocessing is a major source of variance, and conclusions from a single configuration may be unreliable, emphasizing the need for robustness analyses.

Impact of tissue type. In Figure 3, we quantify the impact of tissue type by comparing the performance of models with white and gray matter under identical (dataset, target, task, microstructure feature) configurations. As before, performance is min-max normalized relative to a dummy baseline and we compute

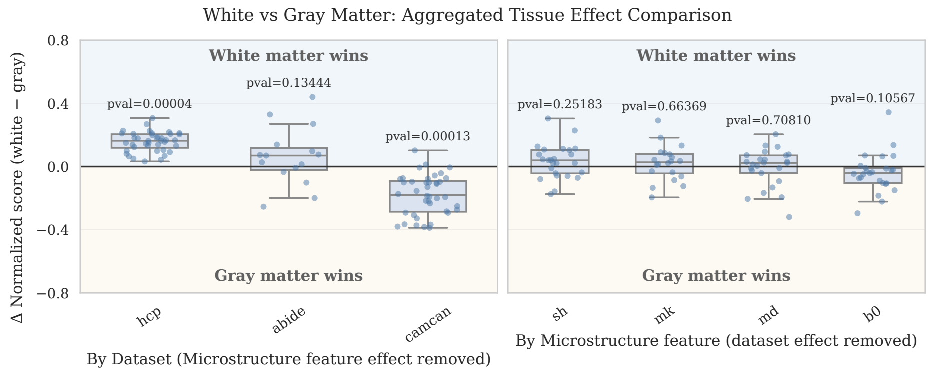

Fig. 3. Fold-level paired differences ($\Delta = \text{white} - \text{gray}$) were computed under identical settings, with significance assessed via a nonparametric paired bootstrap test. *(Right)* Dataset-level tissue effects after removing microstructural feature-specific baselines reveal strong dataset dependence (HCP favors white matter, CamCAN favors gray matter). *(Left)* Microstructure-level effects after removing dataset–task baselines show no consistent tissue preference across representations.

$\Delta = \text{white} - \text{gray}$. To disentangle tissue influence from dataset and representation biases, we residualize the effects in two complementary ways: *(Left)* removing microstructure feature-specific baselines to isolate dataset-dependent tissue effects; *(Right)* removing dataset–task baselines to isolate microstructure feature-dependent tissue effects. Statistical inference is performed on configuration-level mean paired differences to account for fold dependence, using a nonparametric paired bootstrap test to assess whether the mean tissue effect differs from zero.

Figure 3 shows that tissue effects are strongly dataset-dependent. In HCP, positive effect distributions indicate consistent white-matter superiority across configurations. In contrast, CamCAN exhibits the opposite pattern, with gray matter showing a stable advantage. For ABIDE, the distribution of differences is centered around zero, suggesting no clear advantage of either tissue type. Moreover, no microstructural representation demonstrates a consistent tissue preference.

Impact of diffusion microstructure features To assess whether diffusion-derived microstructural features provide predictive value beyond non-diffusion signal, we quantified model-wise improvements relative to b0. The results show that diffusion microstructural information does not uniformly enhance performance across tasks or datasets. For tasks such as gender classification in HCP and CamCAN, diffusion microstructural features do not confer a systematic advantage over b0. It suggests that non-diffusion signal –likely reflecting global anatomical properties such as brain size– already captures most of the discriminative information. In contrast, tasks more directly linked to microstructural variation show measurable gains from diffusion metrics. Age prediction in

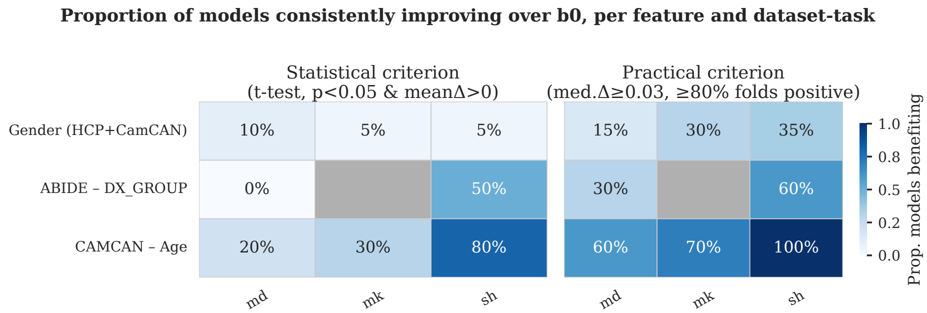

Fig. 4. Proportion of models outperforming the non-diffusion baseline (b0), stratified by task and microstructural feature, based on fold-level normalized differences (Δ = microstructure feature $-$ b0). *(Left)* Statistical criterion (paired bootstrap test, p $<$ 0.05 and mean $\Delta > 0$); *(Right)* Robustness criterion (median $\Delta \geq 0.03$ and $\geq$80% positive folds). Benefits are task-specific rather than universal.

CamCAN and diagnostic classification (DX_GROUP) in ABIDE exhibit consistent improvements for several diffusion representations, particularly under linear models operating on array-based features (e.g., spherical harmonics).

Deep learning models demonstrate positive gains in certain gender classification settings, likely reflecting sensitivity to morphology encoded in image-based representations rather than diffusion-specific contrast. Overall, these findings indicate that the added value of dMRI depends strongly on the biological relevance of the task, the dataset characteristics, and the modeling framework, rather than representing a universal performance boost.

4 Conclusion

In this work, we addressed a central limitation in dMRI prediction: the lack of controlled, comparable evaluation across preprocessing choices, microstructural representations, tissues, models, and datasets. By introducing DiffBench, a unified and open benchmark spanning heterogeneous datasets and tasks, we separate configuration-driven variability from modeling effects. Our analysis demonstrates that performance differences are often as sensitive to representation and preprocessing choices as to model family, clarifying why prior results have been difficult to compare or reproduce. Rather than providing isolated performance gains, DiffBench allows to quantify robustness, generalization, and the true added value of diffusion-derived features. Future work will extend the benchmark to additional datasets and predictive tasks. We also plan to explore other microstructural representations, enable region-level analyses to better localize predictive signal and integrate harmonization methods to better assess cross-dataset generalization We see DiffBench as an evolving infrastructure to support reproducible, cumulative progress in dMRI–based predictive modeling.

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